

The Utility Problem

Hosting Capacity (HC) refers to the ability of the electrical grid to host (connect) additional loads, electric vehicles (EVs), or Distributed Energy Resources (DERs) without compromising power quality, reliability, or necessitating major grid upgrades. HC can be assessed for both loads, including EVs, and for DERs such as solar photovoltaics, wind generation, and battery storage.

The increasing integration of DERs and EVs presents challenges for utilities in evaluating the grid's ability to support these resources without exceeding operational limits. Traditional methods for determining HC often rely on manual studies and worst-case assumptions, which can result in unnecessary interconnection restrictions or costly infrastructure upgrades. By leveraging a data-driven approach, utilities can dynamically assess and visualize HC across the grid in real time, enabling optimized integration while maintaining system reliability.

The two known types of hosting capacity are:

- **Hosting Capacity for Load and Electric Vehicles (EVs)** – This assesses the grid's ability to support additional electricity demand from new loads and EV charging.
- **Hosting Capacity for Distributed Energy Resources (DERs)** – This evaluates how much new generation from DERs can be integrated without exceeding system constraints.

The Data Solutions and Methods

The Awesense Platform offers a scalable, data-driven solution for hosting capacity analysis, serving as a centralized hub for analysts and providing essential tools to efficiently conduct these critical impact assessments.

Hosting Capacity for Load or Electric Vehicles (EVs) - refers to an assessment of the maximum number of new loads and/or EV chargers a feeder, conductors, or transformer can support. Such analysis primarily assesses whether the allowable voltage drop and thermal limits (equipment loading) have been exceeded. Analyses are performed by influx modeling (forecasting) of loads added to the grid until one criterion is exceeded.

Hosting Capacity for Distributed Energy Resources (DERs) - refers to an assessment of how much solar PV, wind, or battery storage the grid can integrate. This analysis checks whether the allowable voltage rise and thermal limits (equipment loading) have been exceeded and/or whether the protection coordination and power quality have been affected. For the analysis, the Distributed Energy Resources (DERs) are added to the grid until one or more criteria are exceeded.

For both cases, the analyzed and calculated capacity is then attached to the asset information in the form of attributes. It can be further visualized on the map (for example, in True Grid Intelligence - TGI).

Examples of the HC criteria, their calculation, and how to obtain data from the Awesense Platform:

Voltage Impact Assessment:

- Goal: Ensure voltage remains within acceptable limits under additional load or DER penetration.
- Method:
 - Adding loads or DERs to the grid and monitoring voltage profiles along feeders to detect voltage rise or drop caused by loads or DERs until the maximum allowable voltage increase or decrease is exceeded.
 - Simulate the impact of increased loads or DER capacity on all feeders.
- Required Data:
 - Grid voltage and current time-series data:
 - Source: AMI meters, sensors (Raptor or others), SCADA, reclosers - baseline voltage levels to compare against modeled values. Current data to determine power flow, which directly affects voltage drops/rises.
 - APIs: `grid_get_datasource()` [SQL API], Export Grid [REST API]
 - Feeder topology and impedance data
 - Source: GIS-based grid model with connectivity data.
 - Includes:
 - Resistance and reactance for conductors and other assets (electrical properties data).
 - Feeder length and conductor type (affecting impedance).
 - APIs: `grid_element` [SQL API], `grid_get_downstream` (or other direction) [REST API]
 - Existing load or DER load and generation profiles
 - Source: Installed EV chargers, solar, wind, and battery storage time-series, consumption data - quantifies new grid power that affects voltage rise.
 - APIs: `grid_get_datasource()` [SQL API]

Thermal Limits:

- Goal: Determine available capacity before thermal limits are exceeded.
- Method:
 - Compare peak loading conditions against equipment thermal limits (based on equipment nameplate ratings).
 - Adding loads or DERs to the grid until transformers and feeders reach their maximum allowable ratings according to their nameplate ratings.
 - Simulate the impact of increased loads or DER capacity on all feeders.

- Required Data:
 - Feeder and transformer load time series. If not available, can be obtained by aggregating time series using a combination of grid tracing and time-series aggregation functions.
 - Transformer kVA ratings (nameplate capacity), cable (line) current carrying capacity.
 - Grid connectivity data (GIS-based asset topology).
 - Existing load or DER generation profiles
 - Source: Installed EV chargers, solar, wind, and battery storage time-series data - quantifies new grid power that affects thermal limits.
 - APIs: `grid_element` [SQL API], `grid_get_downstream` (or other direction) [REST API], `grid_get_datasource()` [SQL API]

Power Quality Assessment:

- Goal: Ensure the integration of DERs, EVs, and new loads does not deteriorate power quality beyond acceptable thresholds (e.g., harmonics, flicker, imbalance, voltage limits), which can affect sensitive equipment, metering accuracy, and overall grid performance.
- Method:
 - Monitor power quality indicators such as overall voltage, Total Harmonic Distortion (THD), voltage flicker, and phase imbalance at key points on the network.
 - Analyze phase loading symmetry, mainly where single-phase EV chargers or rooftop PV systems are installed disproportionately.
- Required Data:
 - Power quality time-series data from smart meters, PQ monitors, or SCADA (e.g., voltage/current harmonics, flicker, THD).
 - Phase assignment data for loads, DERs, and EV chargers (to evaluate imbalance).
 - Feeder topology data.
 - Equipment and component sensitivity thresholds.
 - Source: Meter feeders, SCADA, load flow models or sum of all loads on the feeder - determine the power flow that directly affects the phase unbalance.
 - APIs: `grid_element` [SQL API], `grid_get_downstream` (or other direction) [REST API], `grid_get_datasource()` [SQL API]

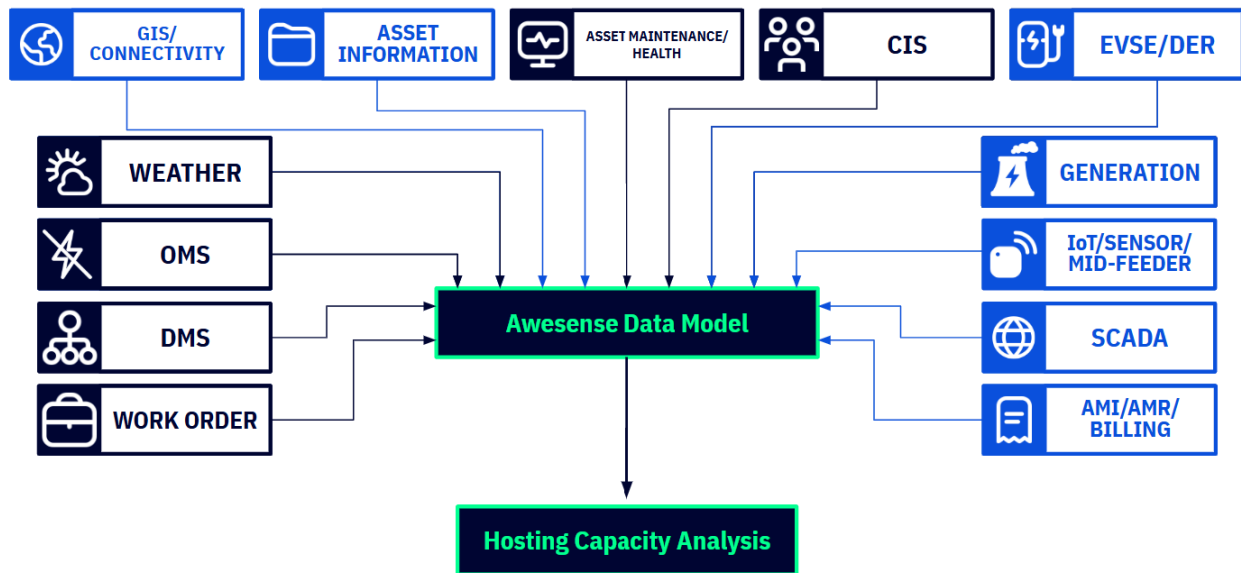
By leveraging the above-mentioned criteria, utilities can determine:

- Where additional DER capacity can be safely accommodated.
- Where grid reinforcement or operational adjustments are needed.
- How to refine interconnection policies to balance load or DER growth and grid stability.

The Awesense Platform centralizes, synchronizes, and ensures seamless access to all necessary data for effective hosting capacity analysis.

Required Data Sets

The required data for this use case are visualized in the following diagram. Data sources highlighted in [royal blue](#) are required for this use case.



Available Use Case Implementations:

Users of the Awesense Platform can access this use case by the following means:

1. Please contact [Awesense](#) if you are interested in the implementation of this use case.